

MIDISTRUCT Variations

Intelligent MIDI Variation Generator for REAPER

by Acrosonus Mastering

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1. What is MIDISTRUCT Variations?

MIDISTRUCT Variations is a REAPER Lua script that analyzes the active MIDI take of a selected item and instantly generates **12 distinct musical variations**, each placed in its own **Fixed Lane** on the same track.

Every variation is produced **non-destructively** — your original (Lane 0) is never modified. The entire operation is undone in a single Ctrl+Z.

One number controls everything: the **Madness Dial** (1–10). At 1, subtle humanization. At 10, radical creative mutation.

Original MIDI item (Lane 0 — unchanged)

```
|
v
[MIDISTRUCT v4]
|
|---> Lane 1  · Syncopation
|---> Lane 2  · Smart Velocity
|---> Lane 3  · Octave & Simplify
|---> Lane 4  · Strummer
|---> Lane 5  · Gate / Staccato
|---> Lane 6  · Power Harmonizer
|---> Lane 7  · Ghost Delay
|---> Lane 8  · Melodic Mirror
|---> Lane 9  · Retrograde
|---> Lane 10 · Ratchet / Stutter
|---> Lane 11 · Reich Phasing
|---> Lane 12 · Chord Shatter
```

Fixed Lanes (REAPER 7+): all variations live on the same track, stacked vertically. Click lane headers to solo/mute. Use REAPER's built-in comp tool to build a composite from multiple lanes.

2. Installation

Requirements:

- REAPER 7.0 or later (Windows, macOS, Linux)
- No external Lua libraries required
- No VST or audio plugins required

Steps:

1. Copy MIDISTRUCT_Variations_v4.lua into your REAPER Scripts folder
 - Windows: %APPDATA%\REAPER\Scripts\
 - macOS: ~/Library/Application Support/REAPER/Scripts/
 - Linux: ~/.config/REAPER/Scripts/
 2. In REAPER: **Actions > Show action list > Load** and select the file
 3. *(Recommended) Assign a keyboard shortcut — Alt+V works well*
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3. Quick Start

1. **Select a MIDI item in the Arrange view**
2. **Press your shortcut (or run from the Action List)**
3. **Enter a Madness level (1–10) and press Enter**
4. **Click lane headers to solo/mute and audition each variation**
5. **Press your shortcut again for a fresh set — Madness is remembered**

Works with any MIDI item including imported .mid files at any PPQ resolution. All note positions are converted through absolute project time to ensure correct timing regardless of the source file's internal resolution (480 PPQ, 96 PPQ, etc.).

4. The Madness Dial

A single number from **1 to 10** scales the intensity of all 12 algorithms simultaneously.

Level	Character	What changes
1	Barely perceptible	Micro-timing drift, gentle velocity shaping
2–3	Subtle humanization	Light syncopation, soft dynamics
4–5	Balanced (default: 5)	Noticeable but musical changes
6	Creative	Ghost Delay pitch drift activates

Level	Character	What changes
7–8	Expressive	Wide strums, octave jumps, strong ratchets
9–10	Radical mutation	80% syncopation, full retrograde, chaotic shatter

Your last-used value is automatically remembered — the dialog pre-fills it on the next run.

5. The 12 Variations

Each variation starts from a **fresh independent copy** of your original notes. Lane names include the Madness level (e.g. **Syncopation [M7]**).

Lane 1 · Syncopation

Family: Rhythm | **Best for:** Bass lines, melodic riffs, drums

Shifts a configurable fraction of notes forward by exactly **half a beat** (one eighth note), creating an off-grid groove. Remaining notes receive micro-timing drift. All displaced notes are boundary-clamped to the item length.

Madness 1	Madness 5	Madness 10
10% displaced, ±3 tick drift	30% displaced, ±10 tick drift	80% displaced, ±30 tick drift

Try this on: A straight 4/4 bass line → instantly sounds like funk or reggae.

Lane 2 · Smart Velocity

Family: Dynamics | **Best for:** Piano, pads, melodic leads

Reshapes dynamics based on each note's position within the beat. Downbeat notes boosted, off-beat notes softened. A **sinusoidal cosine swell** adds natural dynamic breathing. Final humanization layer prevents mechanical uniformity.

Try this on: A flat-velocity piano roll → becomes an expressive dynamic performance.

Lane 3 · Octave & Simplify

Family: Pitch + Structure | **Best for:** Dense arrangements, pads

Three passes: **weighted note removal** (shorter notes more likely culled), **directional octave shift** (below C4 shifts up, above C4 shifts down), **legato-glue** (surviving notes stretched toward next note, with same-pitch collision guard).

Try this on: A dense chord voicing → becomes a sparse, airy melodic fragment.

Lane 4 · Strummer

Family: Rhythm | **Best for:** Chord progressions, stacked notes

Groups simultaneous notes (± 2 tick tolerance) and staggers them like a guitar strum or EDM sweep. Direction alternates per chord for natural picking feel. Organic micro-jitter (± 2 –6 ticks) on every step.

Madness 1	Madness 10
8 ticks between steps, $\times 0.98$ vel decay	40 ticks between steps, $\times 0.72$ vel decay

Try this on: Block piano chords $\rightarrow$ sounds like a strummed acoustic or EDM sweep.

Lane 5 · Gate / Staccato

Family: Rhythm | **Best for:** Pads, strings, slow melodic lines

Cuts every note to a fraction of its original length. Velocity is boosted to compensate for lost sustain. Minimum length floor prevents inaudible micro-notes.

Madness 1	Madness 5	Madness 10
40% length, $\times 1.10$ vel	25% length, $\times 1.20$ vel	8% length, $\times 1.45$ vel

Try this on: A sustained pad $\rightarrow$ becomes an aggressive rhythmic pluck sequence.

Lane 6 · Power Harmonizer

Family: Harmony | **Best for:** Lead synths, bass lines, melodic hooks

Adds a perfect fifth and a sub-bass octave to every note. The fifth is **snapped to the detected scale** of your source material to stay diatonic. Also writes a **CC74 filter riser** (smoothstep cubic curve) for automatic build-up automation.

Layer	Interval	Velocity
Original	± 0 semitones	100%
Perfect Fifth (scale-snapped)	+7 semitones	70–100%
Sub-Bass	–12 semitones	50–92%

Try this on: A thin lead $\rightarrow$ becomes a wide power-chord synth stack.

Lane 7 · Ghost Delay

Family: FX | **Best for:** Leads, arpeggios, drum hits

Creates 2–5 MIDI echo copies per note (Madness-dependent), each softer, shorter, and (at Madness 6+) rising +1 semitone per repeat (tape wobble simulation). Echoes beyond the item boundary are discarded cleanly.

Try this on: A melodic hook $\rightarrow$ becomes a cascading rhythmic bounce. No VST delay plugin needed.

Lane 8 · Melodic Mirror

Family: Pitch | **Best for:** Melody lines, counterpoint composition

Inverts the melody around its **median pitch axis** using the formula **mirrored** = $2 \times \text{axis} - \text{original}$. Out-of-range notes are octave-folded (not clamped) to preserve harmonic interval identity. Duplicate pitches at the same position are deduplicated.

Try this on: A rising melody → generates a descending counter-melody that harmonically fits the original.

Lane 9 · Retrograde (*new in v4*)

Family: Time | **Best for:** Any material — classical counterpoint technique

Plays the melody **backwards in time**. Controls what fraction of the item (from the end) is reversed — notes outside the window pass through unchanged for smooth transitions.

Madness 1	Madness 5	Madness 10
Last 15% reversed	Last 57% reversed	Entire item reversed

Original: A B C D E F G H

M5 Retro: A B C H G F E D (last 57% reversed)

M10 Retro: H G F E D C B A (full reversal)

Try this on: Any melody → creates a new mathematically-related melody. Layer with the original for canon-like counterpoint.

Lane 10 · Ratchet / Stutter (*new in v4*)

Family: Rhythm | **Best for:** Trap hi-hats, synth stabs, IDM glitch

Detects long notes (relative to median length) and subdivides them into rapid-fire bursts. Short notes pass unchanged. Velocity decays across each ratchet step.

Madness 1	Madness 10
8% chance, 2 subdivisions, $\times 0.95$ decay	55% chance, 8 subdivisions, $\times 0.65$ decay

Long note: [=====]

Ratchet M3: [====] [====] [====] [====]

Ratchet M10: [==] [==] [==] [==] [==] [==] [==] [==]

Try this on: A sustained chord → becomes a rhythmic hi-hat pattern. A slow note → transforms into a glitchy IDM stutter.

Lane 11 · Reich Phasing (*new in v4*)

Family: Rhythm | **Best for:** Loops, repetitive patterns, minimalist textures

Relocates notes onto a repeating grid of **N beats** (7 by default, 3 at Madness 10). This creates a polymetric relationship — the pattern drifts against a 4/4 grid and realigns only after **1cm(N)**,

bar) beats. A micro-nudge accumulates per beat to simulate Reich's tape acceleration technique. Pitch and velocity are preserved exactly.

Madness 1	Madness 10
7-beat grid, 4 tick nudge	3-beat grid, 24 tick nudge

Try this on: A short repeating motif → evolves into a hypnotic polyrhythm when layered with the original.

Lane 12 · Chord Shatter (*new in v4*)

Family: Rhythm + Pitch | **Best for:** Chord progressions, piano chords

The aggressive complement to Strummer. Keeps the bass note on the beat; scatters upper chord tones across the rhythmic grid at configurable step intervals. At high Madness, random octave jumps add wide-register harmonic color.

Madness 1	Madness 10
30% chords shattered, half-beat steps, no jumps	95% shattered, 1/8 steps, 50% octave jumps

Try this on: A block chord progression → becomes a melodic arpeggio. At M10 → explodes into a deconstructed wide-register texture.

6. Use Cases by Genre

Genre	Recommended Lanes	Madness
EDM / Big Room	Power Harmonizer + Ghost Delay	7–9
Trap / Hip-Hop	Syncopation + Gate/Staccato + Ratchet	6–8
House / Funk	Syncopation + Smart Velocity + Strummer	4–6
Cinematic	Melodic Mirror + Octave & Simplify + Retrograde	3–5
Ambient / Pad	Gate/Staccato + Octave & Simplify	2–4
Classical / Jazz	Melodic Mirror + Smart Velocity + Retrograde	3–6
Minimalism / IDM	Reich Phasing + Ratchet	5–9
Acoustic Guitar	Strummer + Chord Shatter	4–7
Subtle Humanize	Smart Velocity + Syncopation	1–3

7. Tips & Workflow

Layer original + variation Keep Lane 0 and one variation active. Stack Gate/Staccato over the original for instant rhythmic tension with warmth preserved.

Retrograde + original = canon Play Lane 0 and Lane 9 together. At M3–5, the reversed section creates a brief tail that sounds like composed counterpoint.

Reich Phasing as a send Route Lane 11 to a reverb bus at lower volume. The polymetric drift creates a natural evolving texture without any automation.

Re-run on a variation Duplicate the MIDI item from one lane, make it active, then run MIDISTRUCT again. You get variations of variations — great for finding unexpected material.

Ratchet on drums Apply MIDISTRUCT to a MIDI drum pattern. Lane 10 subdivides hi-hat notes into 32nd-note flams automatically.

Chord Shatter then Strummer Run on your chords for Lane 12 (Shatter). Use the shattered result as input for a second MIDISTRUCT run. Lane 4 (Strummer) on that result adds micro-timing on top of the dispersion.

8. Important Notes

- Only the **first selected item** is processed — no batch mode
 - **Lane 0 (original) is never modified** — all lanes are independent copies
 - **Ctrl+Z undoes all 12 lanes in one operation**
 - **Each run is unique** — high-entropy RNG seed, two runs at the same Madness produce different results
 - **Works with any PPQ resolution** — imported `.mid` files handled correctly via absolute time conversion
 - **REAPER 7.0+ required for Fixed Item Lanes**
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9. Troubleshooting

“No media item selected” Click on a MIDI item in the Arrange view before running.

“Active take is not MIDI” Right-click the item → Take → select the MIDI take.

Lanes do not appear Right-click the track header → Show lanes, or drag the track height down to reveal them.

Ratchet or Chord Shatter do nothing Ratchet needs long notes (relative to median) to trigger. Chord Shatter needs simultaneous chord notes. On monophonic material with uniform note lengths, both are nearly invisible by design.

Reich Phasing sounds identical to original The polymetric effect requires a loop or repeating pattern to drift against. Try M6+ and layer Lane 11 with Lane 0 — the phasing emerges over several bars.

Retrograde only reverses a small section At low Madness, only the last 15% is reversed by design. Increase Madness for a larger reversed window.

Variations sound too similar Increase Madness. Try 7–9 for dramatic results.

Variations sound too extreme Lower Madness. Level 2–3 gives subtle, production-ready humanization.

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